

Online Streaming Platform- Flick's Cloud Architecture and Data Pipeline

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Introduction

The emergence of streaming services that appeal to a wide range of consumers has caused a dramatic change in the digital entertainment market in recent years. Flick is a forerunner in this field, revolutionizing the way consumers engage with and perceive material. Flick is on a mission to transform entertainment by delivering a smooth, customized streaming experience by fusing state-of-the-art cloud architecture with sophisticated data analytics.

With a huge collection of international films, TV series, and unique original material, Flick's platform has something to suit every taste. It uses machine learning and real-time data processing to curate suggestions, making sure users get engaging and inspiring tales. Flick places a premium on high-quality streaming at any time and from any location, whether users are watching content on mobile devices or home systems.

This article examines Flick's creative cloud architectural design, which serves as the foundation for its capacity to handle enormous volumes of data, provide individualized experiences, and sustain reliable performance at scale.

Mission

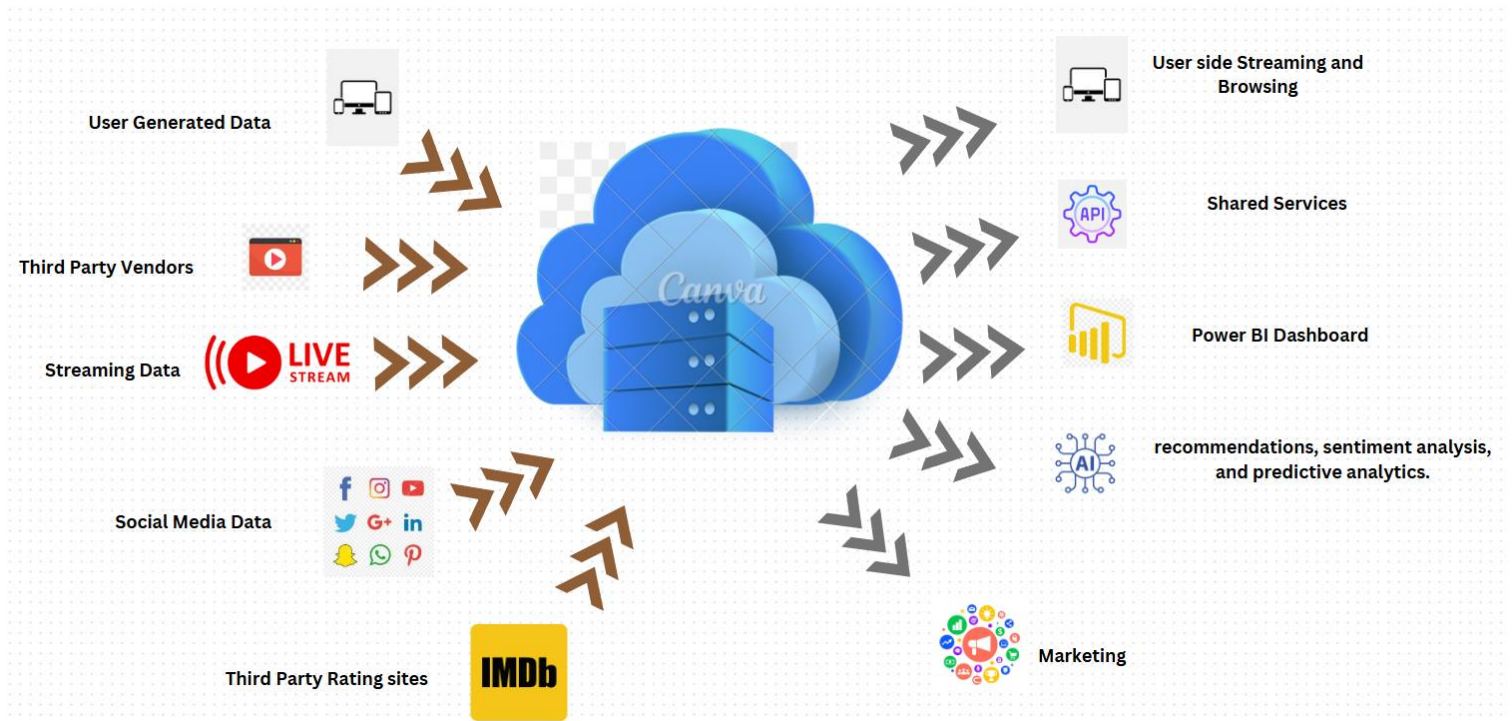
"Revolutionizing entertainment with a personalized, accessible platform that connects global audiences to inspiring and engaging stories."

Objective

1. Content Diversity and Quality
2. Personalized User Experience
3. Seamless Streaming Across Devices
4. User Engagement and Satisfaction
5. Global Accessibility

Vision Diagram

The vision diagram represents the holistic view of the Flick platform's operations, detailing the interplay of various components like data acquisition, processing, and service delivery. Visual illustrations can be included for better comprehension.



The provided diagram illustrates a comprehensive flow of data from its sources through the cloud architecture and into actionable services.

Data Source

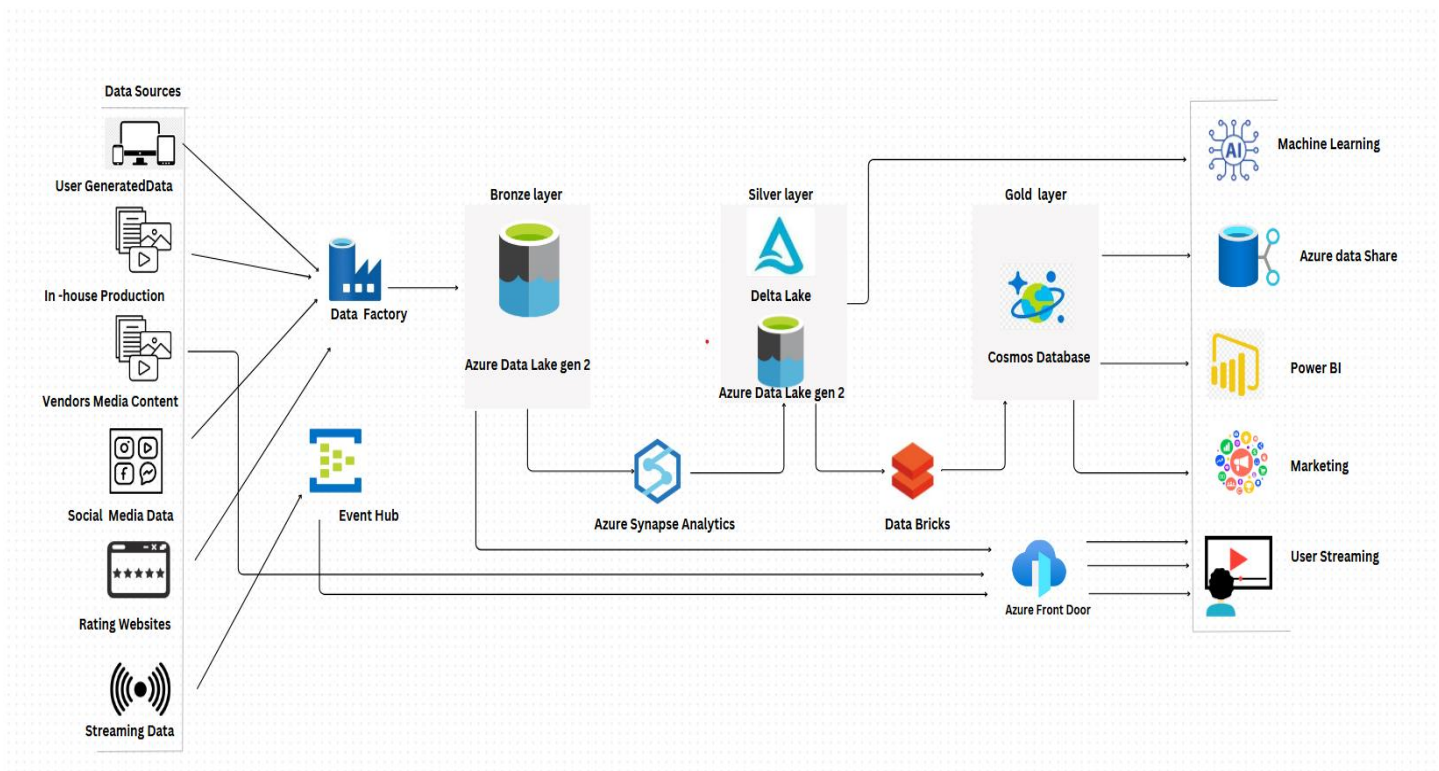
- **User Interaction Data:** Captures user behavior, such as browsing history, search preferences, and streaming habits.
- **Content Metadata:** Provides detailed information about movies, TV shows, and other media assets, including genres, descriptions, and popularity metrics.
- **Streaming Logs:** Tracks session details, quality metrics, and buffering rates.
- **External APIs:** Supplies supplementary data, such as trending topics, reviews, and global audience insights.
- **In-House Production Videos:** We have our own production house from there also we store the videos.

Data Sinks

After processing, the refined data supports several high-value services, including:

1. **End User:** Streaming on various devices without any lagging and high-quality videos.
2. **Machine Learning / AI:** Enhances user experience by providing tailored content suggestions and insights.
3. **Marketing:** Utilize analytics to optimize campaigns, track user behaviour, and improve customer retention strategies.
4. **Business Intelligence (Power BI dashboard) :** Provides actionable insights on content performance, user engagement, and platform growth.
5. **Shared service:** Facilitates partnerships and extends the platform's ecosystem by enabling data sharing with trusted partners.

Flick's Cloud Architecture



The Lakehouse architecture for Flick is shown above. The data is ingested as a bronze layer in Azure Data Lake Gen 2 via Data Factory. As was already indicated, the data is gathered from a variety of sources. Three-layer security, such as the Bronze, Silver, and Gold Layers, are commonly used in architecture.

Data Ingestion Layer

This layer ingests raw data from all sources and streams it into the system.

a) Azure Data Factory

- **Purpose:** A fully managed orchestration tool for collecting, transforming, and loading data (ETL/ELT processes).
- **Use Case:** Handles batch data ingestion for metadata, user actions, and social media data.

b) Event Hub

- **Purpose:** A real-time event ingestion service for streaming data.
- **Use Case:** Processes live streaming data, such as playback events and network telemetry,

ensuring minimal latency.

Data Storage Layer (Lakehouse)

The storage system is divided into three layers, each serving a distinct purpose:

a) Bronze Layer (Raw Data Storage)

- **Service:** Azure Data Lake Gen 2
- **Purpose:** Stores raw data in its original format (structured, semi-structured, and unstructured).
- **Use Case:** Acts as a source of truth and a fallback for reprocessing.

b) Silver Layer (Cleaned & Processed Data)

- **Service:** Azure Data Lake Gen 2 & Delta Lake
- **Purpose:** Processes raw data by cleaning, filtering, and organizing it into a structured format.
- **Features:** Delta Lake supports ACID transactions, versioning, and schema enforcement.
- **Use Case:** Prepares data for analytical queries and insights.

c) Gold Layer (Business-Ready Data)

- **Service:** Azure Cosmos DB
- **Purpose:** Stores high-quality, aggregated data ready for direct consumption by business and analytics systems.
- **Use Case:** Powers dashboards, reports, recommendations, and other consumer-facing applications.

Data Processing Layer

a) Azure Synapse Analytics

- **Purpose:** A unified platform for big data processing, analytics, and querying.
- **Use Case:** Enables complex data aggregations, joins, and transformations on the silver data.
- **Integration:** Reads data from the silver layer and writes insights back into the gold layer.

b) Azure Databricks

- **Purpose:** A collaborative, cloud-based data engineering and machine learning platform.
- **Use Case:**
 - Performs advanced data cleaning and model training for recommendations, churn prediction, or trend forecasting.
 - Supports exploratory analysis for deeper insights.

Data Consumption Layer

a) Machine Learning

- **Purpose:**
 - Delivers predictive analytics (e.g., recommending similar content to users).
 - Enables real-time decision-making for operations like adaptive streaming quality.

b) Azure Data Share

- **Purpose:** Securely shares data with external vendors or stakeholders for collaboration.
- **Use Case:** Allows content providers to access viewership insights for their licensed content.

c) Power BI

- **Purpose:** Provides interactive dashboards for visualizing KPIs, such as user engagement, content performance, and revenue.

d) Marketing

- **Purpose:** Uses insights to personalize promotions, recommend new releases, and optimize ad placements.

e) User Streaming

- **Purpose:** Enables efficient delivery of content to users with low latency through Azure Front Door.

Scalability and Global Delivery

a) Azure Front Door

- **Purpose:** Acts as a global content delivery network (CDN) and load balancer for seamless user experiences.
- **Use Case:** Ensures quick delivery of content and high availability, even during peak demand.

Advantages of This Architecture

1. **Unified Data Platform:** Combines raw, processed, and business-ready data into a single system.
2. **Scalability:** Azure services ensure horizontal and vertical scalability as data volume grows.
3. **Real-Time Analytics:** Event Hub and Synapse Analytics facilitate real-time decision-making.

4. **Cost Efficiency:** Tiered data storage (bronze, silver, gold) optimizes costs by storing raw data cheaply and using more expensive storage only for processed data.
5. **Security and Compliance:** Azure Data Lake and Cosmos DB ensure secure data storage and access control.
6. **Advanced Insights: Integration with Databricks and Azure ML provides AI/ML capabilities for predictive analytics and personalization.**

Conclusion

This Azure-based Lakehouse architecture is designed to handle the complex data processing requirements of a streaming platform. It ensures reliable data ingestion, advanced analytics, real-time processing, and efficient delivery, enabling the platform to enhance user experience, improve operational efficiency, and scale with demand.